

Financial Risk Analysis

JP Morgan

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Tasks



Data Cleaning and Formatting



Descriptive transactional analysis



Customer Profile Building



Financial Risk Identification



Visualisation



Hypothesis Testing

Task Summary



What each task does?

1. Descriptive transactional analysis

This task analyzes overall transaction behavior across accounts and time periods.

It focuses on calculating summaries such as credits, debits, net inflow, and identifying trends or inactive accounts.

The goal is to understand transaction patterns and account performance using descriptive statistics.

2. Customer Profile Building

This task groups customers/accounts based on transaction frequency, balances, and activity levels

It create customer segment such as high-net inflow accounts or low-balance high-frequency users.

The goal is to build customer profiles that help understand different banking behaviors.

3. Financial Risk Identification

This task identifies risky or suspicious financial activity within customer accounts.

It includes detecting overdrafts, abnormal balance volatility, and unusual transaction behavior using statistical methods.

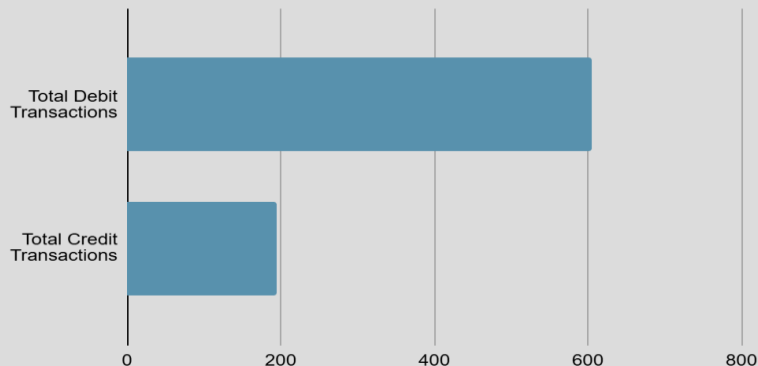
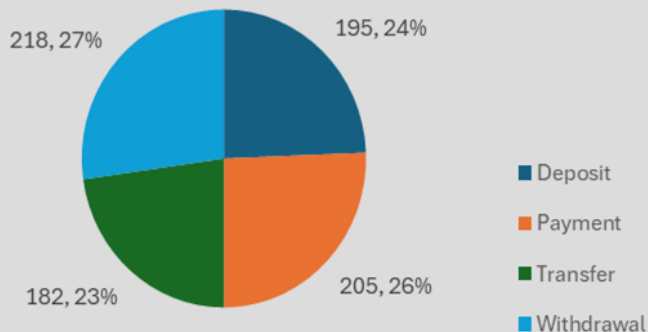
The goal is to highlight accounts that may require further investigation.

4. Visualisation

This task visualizes transaction and customer behavior using charts and graphs to identify trends, patterns, and insights more clearly.

5. Hypothesis Testing

This task statistically tests whether different customer segments behave differently based on financial metrics.



190

No. of Unique customer

There were total 190 different customers

800

Total Transaction

There were total 4 different type of transaction

\$51,086

Average transaction

On average each transaction was of 50k+

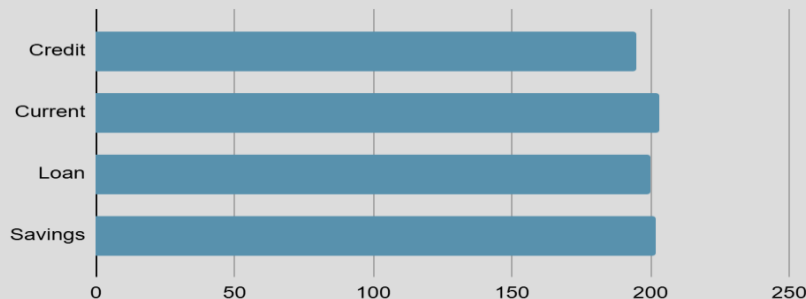
194

No. of Accounts

There were total 194 different account of 4 types

- As we can see that the majority of the payments were of type Withdrawal
- We can also see that we had a total of 800 transactions done by 194 accounts which were owned by 190 customers
- One of the most important thing that we can see here is the total rabbit transactions were around 600 but the credit transaction was just below 200 Making the net inflow negative Which we will see later

Transaction Count vs. Account Type



0.4681

Average Risk Score

The average risk score
across all transaction

50255

Average Debit Amount

Average Transaction amount
of Outflow transaction

53665

Average Credit Amount

Average Transaction amount
of inflow transaction

4

Average No. transaction

On average each Account
did 4 transactions

576

Average Credit Score

-1,99,39,833.10

Net Inflow

387

High Risk Transaction

There were total 387 High
risk transaction out of 800

- We can see that current account made the highest number of transaction out of all the four accounts
- We can also see that average credit amount was more than the average debt amount
- Still the net inflow was a negative because the number of credit transactions were Only approx. 200 out of 800 total transactions
- Another important thing is there were around 387 high risk transaction who had a risk score of above average

Task 2

Descriptive transactional analysis

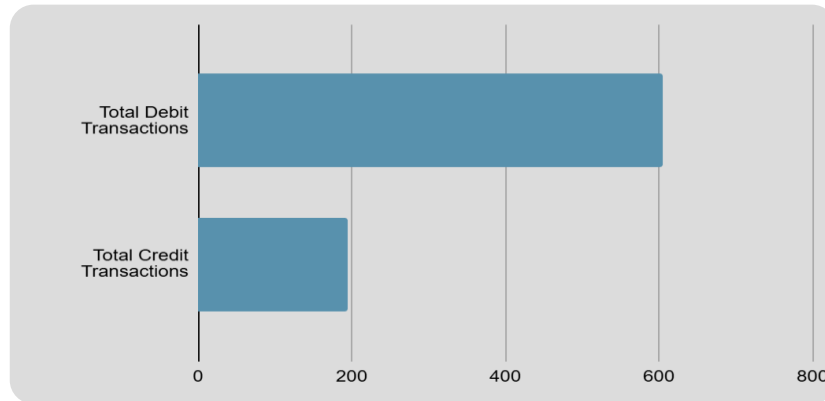
- Calculate monthly and yearly summaries of total credits, debits, and net transaction volume.
- Plot trends in total credits vs. debits over time.
- Identify top and bottom performing accounts based on net inflow.
- Identify and flag accounts as dormant or inactive if there is a gap of two months or more between consecutive transactions.

Task 2

Key takeaways

Top 3 Accounts by Net Inflow

ACC22036	291395.48
ACC74631	253027.51
ACC54589	225122.50



-1,99,39,833.10

Net Inflow

The net Inflow was negative as there were more debit transaction

3,04,04,566

Total debit Transaction

Sum of Total debit transaction

1,04,64,733

Total Credit Transaction

Sum of total credit Transaction

183

No. of Dormant account

There were total 183 dormant account out of 194

- During the descriptive analysis again we found out that The total debit transaction sum was way higher than total credit transaction sum which caused a negative net inflow
- we also found out that hour of total 194 accounts 183 were dormant that means that the last Two transaction Had a gap of more than 60 days
- We can also see the top three accounts With the highest net inflow

Task 2

Key takeaways

0.4692

Average debit Risk score

Average Risk Score of all
Debit transaction

0.4746

Average credit Risk score

Average Risk Score of all
credit transaction

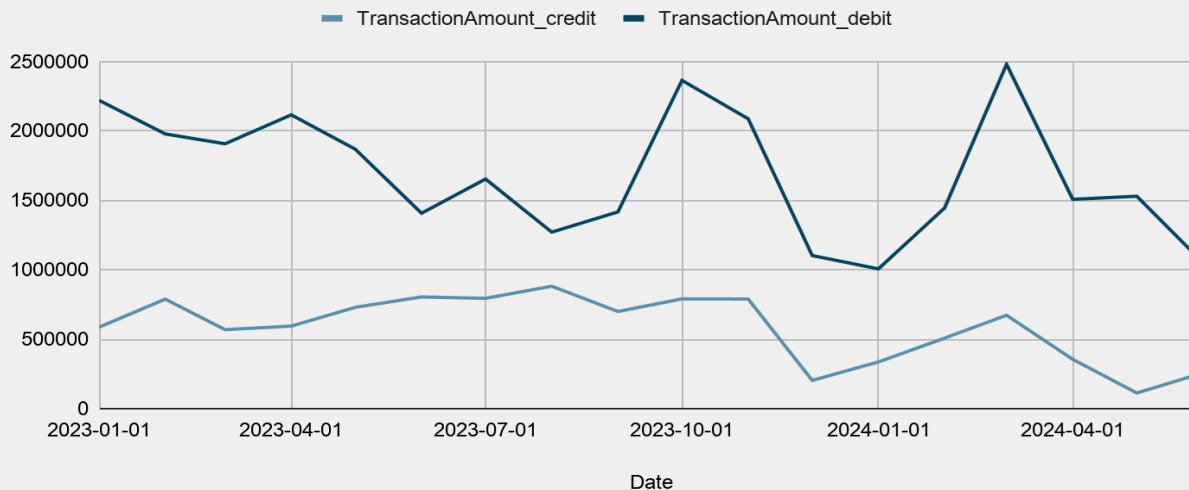
195

No. of Credit Transaction

No. of credit Transactions

- We discovered that the average risk score for the credit transactions were a little higher than average risk scores for debit transactions
- We can also see Out of 800 total transactions 195 only were The credit transactions

Total Transaction credit and Total Transaction debit by Date



Task 3

Customer Profile Building

- Group accounts by activity levels: High, Medium, Low based on transaction frequency on your analysis
- Segment customers by average balance and transaction volume.
- Identify top and bottom performing accounts based on net inflow.
- Create profiles for:
 - High-net inflow accounts
 - High-frequency low-balance accounts
 - Accounts with negative or near-zero balances

Task 3

Key takeaways

34

High Activity Account

Accounts with Average transaction gape of 0-1 month

47

Medium Activity Account

Accounts with Average transaction gape of 1-2 months

113

Low Activity Account

Accounts with Average transaction gap of 2 or more months

29

High Net Inflow Account

Accounts with High Net inflow

2

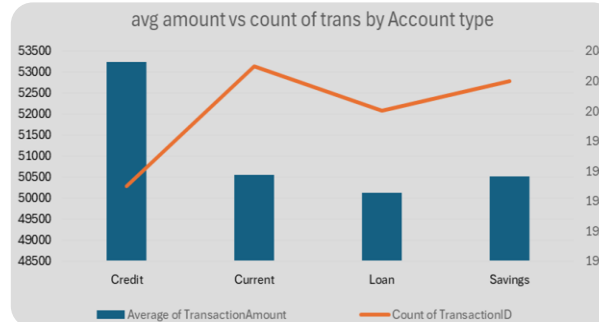
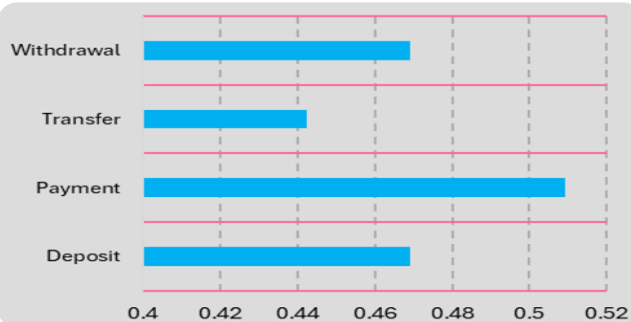
High Freq. Low Balance account

High frequency with low balance accounts

2

Near Zero / Zero balance account

Accounts with Balance Near Zero or Zero



- As we can see that we had 113 low activity counts Which were way higher than Medium and high activity account
- We can also see that there was just only 29 high net inflow accounts
- And there were only two accounts with near zero or zero balance And only to account with high frequency and low balance
- We can also see important thing in the chart it shows that The number of transactions for Current account were way higher Then any other count type but the average transaction balance was really low
- On another hand we can see that the number of transactions for credit accounts were really low But the average transaction amount was the highest out of all
- We can also see the dirt is scores for Payment type of transaction was the highest

Task 4

Financial Risk Identification

- Track accounts with frequent large withdrawals or overdrafts.
- Calculate balance volatility using standard deviation or coefficient of variation.
- Use IQR or z-score methods to detect anomalies.
- Highlight customers with irregular or suspicious transaction behavior.

Task 4

Key takeaways

40

High Risk Account

Account with large withdrawal or overdraft balance

10

Abnormal Transactions

Abnormal Transactions using IQR

99

Suspicious / Irregular Customer

Account with Overdraft, Large Transaction and irregular CV

384

The Transactions who had higher than average Risk Score

High Risk Transactions

Risk Score by Acc. Types



Avg credit score by acc type



- During risk analysis we saw that there were 40 high risk account with who had large withdrawal or overdraft balances
- there were Ten abnormal transactions that were statistically Defined using IQR
- We can also see out of 800 transaction again 384 were high risk transactions
- Again we can see that the Chris score for the payment type of Transaction was highest
- And we can also see the account Of credit type at the highest credit score And saving account at the lowest credit score

Task 5

Visualisation

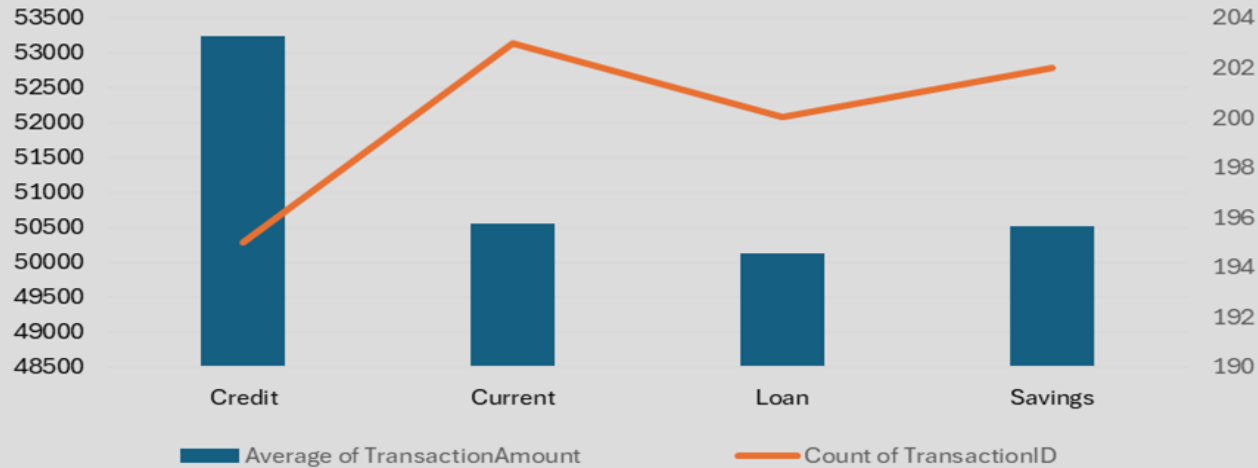


Conduct extensive exploratory data analysis with attractive visualizations for your findings

Task 5

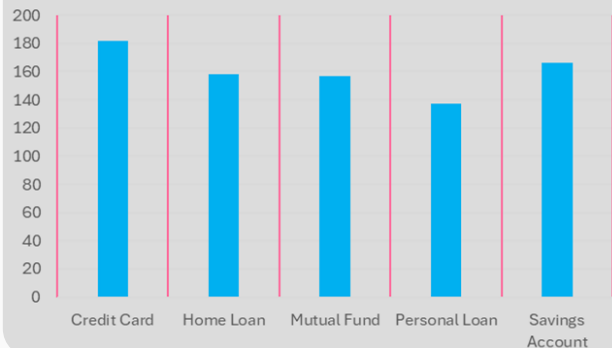
Key takeaways

avg amount vs count of trans by Account type

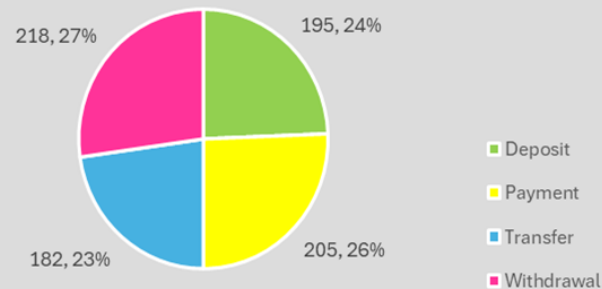


- Here we can see again that Transaction amount on average was highest for credit types of account With the number of transactions being the lowest
- You can also see The most transactions were made for the credit cards Prototype
- And withdrawal was The high Number of transaction type

No of Trans. by Product by Acc type



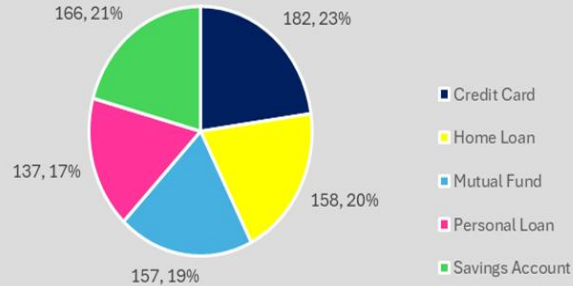
No. of Transaction by Trans. Type



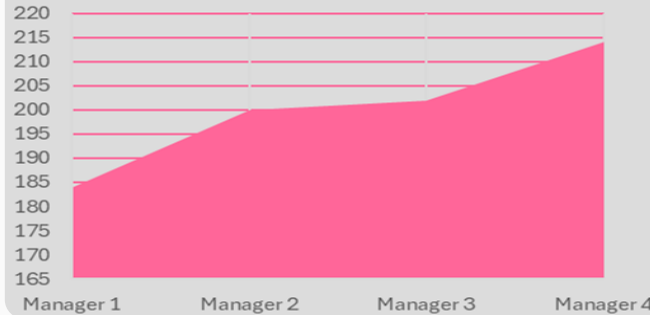
Task 5

Key takeaways

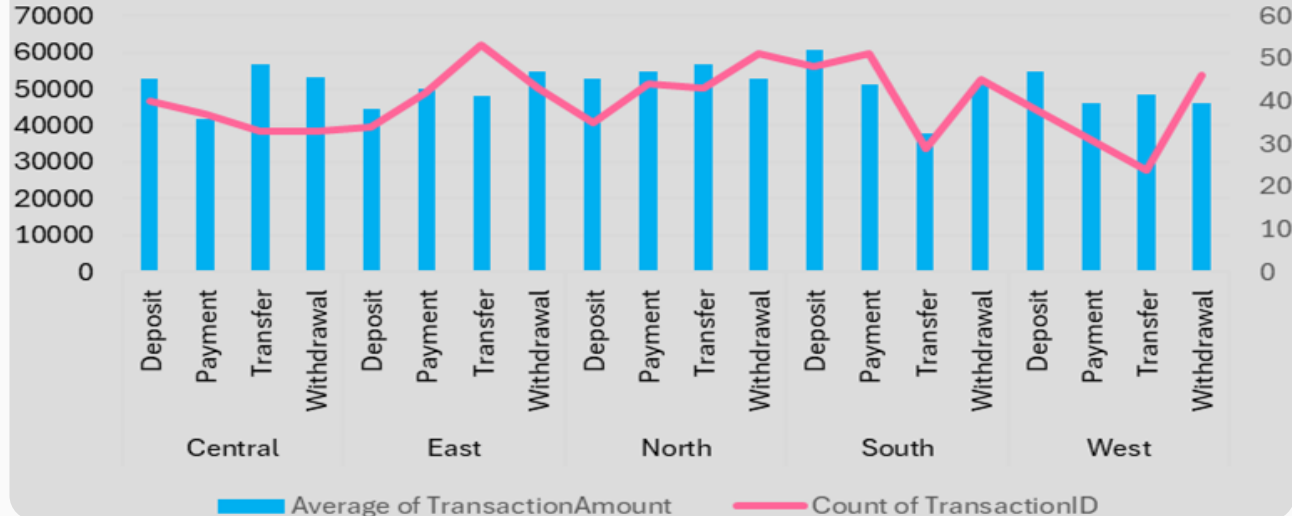
No. of Transaction by Product Type



Transaction No. by Manager



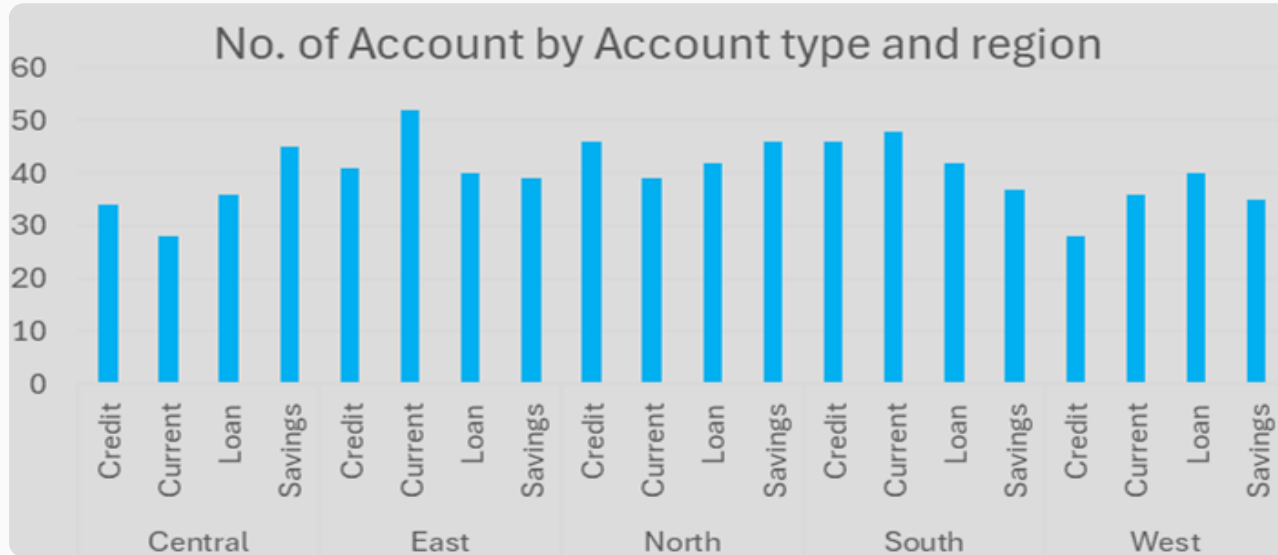
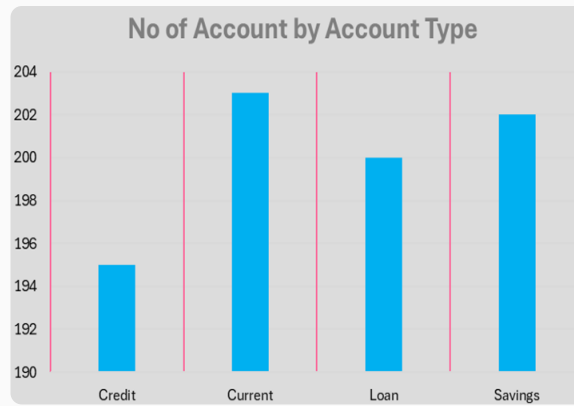
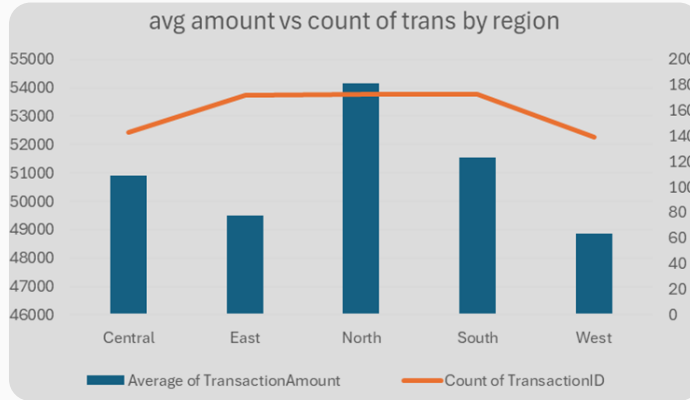
Avg trans. vs No. of Trans. by Acc. type and region



- Here we can see that the manager 4 was handling the highest number of trans But later will you find out that the average transaction amount handled by a manager for was the lowest
- So manager 4 was handling low amount transactions
- Again we can see credit Card was the highest product that the Transactions were made for

Task 5

Key takeaways



- Here we can see the Accounts were of type current followed by savings. We can also notice a big thing that Even though there were low number of current account the average transaction amount was the highest.
- We can also see The Transaction amount on N was higher than Any Region.

Task 6

Hypothesis Testing

- Test whether high-volume transaction accounts have statistically higher average balances than low-volume accounts.
- Conduct hypothesis testing based on segmentation.

Hypothesis

Null Hypothesis	Result
There is no statistically significant difference in average transaction amount between high-balance and low-balance accounts.	True
There is no statistically significant difference in average risk score between high-transaction and low-transaction accounts.	True
There is no statistically significant difference in average transaction amount between Savings and Current accounts.	True
There is no statistically significant difference in average risk score between the Central region and the other regions.	True
There is no statistically significant difference in average transaction amount between Loan/Credit accounts and Current/Savings accounts.	True
There is no statistically significant difference in average risk score between long-tenure and short-tenure accounts.	True

Key Findings

1

There were total 800 transactions done by one 94 accounts handled by 190 customers

Are you these 194 accounts most accounts were of current account type

Out of 194 accounts 183 were dormant and 113 were low activity account

1

Out of 800 transactions only 195 transactions were of credit Causing a negative net inflow

The average amount for credit transactions were higher than debit transactions But due to high number of debit transactions negative net inflow was maintained

3

Out of 800 transaction 387 were high risk And out of 194 accounts 14 were high risk accounts And out of 190 customers 99 customers were suspicious or irregular

The account type of credit had the lowest number of transactions but average transaction amount was highest

4

Out of all the regions North region Had the highest average transaction amount

Highest number of transactions were made from current account type

The product which most transactions were made for Was credit card

5

After hypothesis we also saw that there was statistically no To little difference between the segmentations of Customers, Account And regions

manager 4 was handling the highest number of transactions but the average transaction amount was the lowest

Video - <https://drive.google.com/file/d/1DSQLgyN6JOgNbZFTBSVBh0lcf2QMWSiP/view?usp=sharing>



Thank you!